Qn. Set Code-7

Semester: 5th
Programme: B.Tech
Branch: CSE

AUTUMN END SEMESTER EXAMINATION-2024

5th Semester B.Tech

DATA MINING AND DATA WAREHOUSING

CS30013

(For 2023 (L.E), 2022 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Compute the Euclidean distance between two data objects represented by the tuples (22, 1, 42, 10) and (20, 2, 38, 8).

(b) If a classifier correctly classifies 70 tuples out of 110 tuples of test data, what will be the error rate of the classifier?

(c) Differentiate between median and midrange of a data set.

(d) Define similarity and dissimilarity matrix with suitable example.

(e) Distinguish between noise and outliers.

(f) State the use of confusion matrix.

(g) Define with suitable example what do you mean by correlation analysis.

(h) What will be the Mean Square Error for the data set?

Y	5	6	7	8	3
Y'	5	4	7	9	2

(i) How to improve the performance of the Apriori method for association mining?

(j) Does SVM algorithm have high classification accuracy in high dimensional space? Justify your answer.

SECTION-B

2. (a) Differentiate between OLTP and OLAP. State at least six differences. [5]

(b) Explain the process of knowledge discovery in databases. [5]

3. (a) Consider the following transactions: [5]

Tid	Items
1	a, b
2	b, c, d
3	a, c, d, e
4	a, d, e
5	a, b, c
6	a, b, c, d
7	a, b, d
8	b, c, e

Construct the FP Tree. Show the tree separately after reading each transaction.

(b) What is clustering? Differentiate between Hierarchical and Partitional clustering. [5]

4. (a) Describe the k-means and k-medoids algorithm in terms of shape of cluster that can be determined and parameter that can be specified. Illustrate the strength and weakness of k-means in comparison with k-medoids. [5]

(b) Solve the following problem using k-means algorithm. [5]

Consider 10 points {X1, X2, X3, X4, X5, X6, X7, X8, X9, X10} with the following coordinates into two clusters:

X1(1, 1), X2(1, 3), X3(2, 2), X4(3, 1), X5(4, 4), X6(8, 1), X7(8, 3), X8(9, 2), X9(10, 1), X10(10, 3).

Suppose initial cluster center for each cluster are X1, X5 respectively. Use k-means algorithm to show the clusters.

i. The two cluster centers after the first round of execution.

ii. The final two clusters.

5. (a) What is neural network? Explain the feed-forward and back-propagation of error with a suitable example. [5]

(b) Consider the given credit audit database. Predict the "Response" of the customer "David" using K-nearest neighbors algorithm (assume k=3). [5]

Customer	Age	Income	No. of Credit cards	Response
John	35	35k	3	NO
Rachel	22	50k	2	YES
Hannah	63	200k	1	NO
Tom	59	170k	1	NO
Nellie	25	40k	4	YES
David	37	50k	2	?

6. (a) Predict the value of Y if X=15 using linear regression. [5]

X	2	4	6	8
Y	3	7	5	10

(b) Suppose a hospital tested the age and body fat data for 11 randomly selected adults with the following result: [5]

Age	%fat
23	9.5
23	26.5
52	34.6
54	42.5
54	28.8
56	33.4
57	30.2
58	34.1
58	32.9
60	41.2
61	35.7

i. Calculate the mean, median, and standard deviation of age and %fat.

ii. Calculate the correlation coefficient (Pearson's product moment coefficient). Are these two variables Age and %fat positively or negatively correlated?

7. (a) Consider a model that can predict either a person has the disease or not from a sample dataset of total 100 samples. Calculate (i) Accuracy, (ii) Recall, (iii) Precision and (iv) F1-Score. [5]

	Actual: YES	Actual: NO
Predicted: YES	22	10
Predicted: NO	8	60

(b) Suppose a group of 12 sales price records has been sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into 3 bins by each of the following methods: [5]

i. Equal frequency partitioning

ii. Equal width partitioning

For each type of partitioning, smooth out the bin values by **boundary**.

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